

Wheeler Memory

Project Darman

A memory system that remembers like you do — imperfectly, associatively, and influenced by context.

The Problem

Why Memory Matters in LLMs

- **Confabulation:** Current LLMs hallucinate confidently with zero uncertainty signals
- **Statelessness:** No memory = no personalization, no context persistence across sessions
- **Information Loss:** Context windows close; conversational history is discarded
- **Enterprise Risk:** Mission-critical applications can't rely on models that don't know what they don't know

Today's LLMs are amnesiacs who don't admit it.

Market Opportunity

The \$100B+ LLM Market Needs Memory

Market Segment	Size	Pain Point
Enterprise Copilots	\$30B+	Need reliable recall for customer history, docs
AI Search/RAG	\$20B+	Vector DBs don't reconstruct; retrieval $\neq$ memory
Agentic Systems	\$25B+	Agents need persistent state, forgetting curves
Personalized LLMs	\$25B+	User models require associative recall

Memory is the missing layer. Every major LLM platform (OpenAI, Anthropic, Google) is exploring it. The first production-ready system wins.

Our Solution: Wheeler Memory

A Cellular-Automaton Memory Engine

Text → CA Attractor → Correlation Search → Context-Dependent Reconstruction

- **Unique:** Uses 3-state cellular automaton instead of vectors
- **Reconstructive:** Every recall is context-dependent (like human memory)
- **Temperature-Aware:** Explicit confidence tiers prevent confabulation
- **Local-First:** Runs entirely on-device; no cloud dependency
- **GPU-Optional:** Works on CPU; accelerates on HIP/ROCm/CUDA

"Darman doesn't retrieve. Darman reconstructs."

How It Works (Simplified)

```
Text Input
  ↓
SHA-256 hash (or semantic embedding)
  ↓
64×64 grid seed
  ↓
3-State CA Evolution (40-100 ticks)
  ↓
Stable Attractor Pattern
  ↓
Search: Pearson correlation against query
  ↓
Reconstruct: Blend stored + query, re-evolve
  ↓
Result: Context-dependent memory
```

Convergence: ~3 ms on CPU. Deterministic. Reproducible.

Key Differentiator #1: Reconstructive Recall

Darman Reconstructs, Not Retrieves

Same stored memory reconstructs differently depending on context:

```
blend = (1 -  $\alpha$ ) × stored +  $\alpha$  × query    ( $\alpha = 0.3$ )  
reconstructed = evolve_and_interpret(blend)
```

- Query "machine learning" vs. "debugging" → different reconstructions
- **Like human memory** (Elizabeth Loftus, cognitive psychology)
- **Prevents database-like rigidity** — memories stay alive and contextual

Competitors (vector DBs, RAG): Static retrieval. Same result every time. No context dependency.

Key Differentiator #2: Temperature System

Epistemic Humility — Confidence Built In

```
temp = base_from_hits × decay_from_time  
Half-life: 7 days
```

Tier	Temp Range	Darman's Language
Hot	≥ 0.6	"I clearly remember..."
Warm	≥ 0.3	"I think we discussed..."
Cold	< 0.3	"I vaguely recall, but I'm uncertain..."

- **Prevents confabulation:** LLM can't sound confident about stale memories
- **Competitive advantage:** Customers know when to trust the system
- **Explainable AI:** Confidence is auditable, not a black box

Key Differentiator #3: Local-First, Privacy-Preserving

No Cloud. No API Keys. No Data Escape.

- **All computation on-device:** CPU or local GPU
- **All storage local:** `~/ .wheeler_memory/` by default
- **No vendor lock-in:** Sentence-transformers runs locally; Ollama serves LLM locally
- **Enterprise-ready:** HIPAA, GDPR, CFAA compliance by architecture

Market advantage: Enterprises demand data sovereignty. We deliver it.

Technical Validation

Proof Points

- **167+ automated tests** covering CA dynamics, recall, reconstruction, temperature decay
- **3 major datasets:** MBPP (code), SWE-bench (software engineering), BABILong (long-context reasoning)
- **Diversity validation:** Rotation retry ($0^\circ/90^\circ/180^\circ/270^\circ$) ensures stability across seed angles
- **Reconstruction demo:** Semantic recall on embedding-based paraphrases
- **GPU benchmarks:** HIP/ROCm + CUDA with auto-fallback

All tests reproducible. All benchmarks public.

Architecture: 19 Modules, Deep Stack

```
graph TD; A[Agent Loop (Ollama integration)] --> B[Chunking & Routing (6 domain-specific stores)]; B --> C[Reconstruction Engine (context-dependent blending)]; C --> D[Temperature Dynamics (decay + confidence tiers)]; D --> E[Storage & Recall (Pearson correlation)]; E --> F[3-State CA Dynamics (local max/min/slope rules)];
```

Agent Loop (Ollama integration)
↓
Chunking & Routing (6 domain-specific stores)
↓
Reconstruction Engine (context-dependent blending)
↓
Temperature Dynamics (decay + confidence tiers)
↓
Storage & Recall (Pearson correlation)
↓
3-State CA Dynamics (local max/min/slope rules)

- **Modular:** Swap any layer without touching others
- **Testable:** Each module has isolated test suite
- **Extensible:** Custom embeddings, hashing, GPU kernels

Engine Philosophy

"The Engine Is the Mind. The LLM Is the Voice."

Key insight: Swap the underlying model (GPT-4 → Mistral → local 7B) and Darman's *behavior* stays the same. Only *phrasing* changes.

Why this matters:

- Behavior is deterministic, auditable, provable
- Not dependent on model capability
- Cheaper to operate at scale (smaller models work)
- Personality emerges from memory, not from prompt engineering

Enterprise value: Darman's identity is stable regardless of LLM upgrade.

Design Principles

7 Axioms That Drive Everything

1. **Engine is the mind** — behavior from CA, phrasing from LLM
2. **Reconstructive recall** — context-dependent, not static lookup
3. **Temperature is epistemic humility** — uncertainty is computed, not guessed
4. **Memories are suggestions** — LLM can disagree with stale memories
5. **Minimize LLM dependency** — engine handles search/decay/reconstruction
6. **Local-only by default** — cloud is optional, not required
7. **Formula is the foundation** — CA rule is load-bearing; UI is commentary

These principles prevent feature creep and ensure the system stays coherent as it scales.

Competitive Positioning

Wheeler vs. Alternatives

Aspect	Vector DB	RAG	Wheeler Memory
Retrieval	Static	Static	Context-Dependent
Confidence Signal	None	None	Temperature Tier
Reconstruction	No	No	Yes
Local-First	Sometimes	Sometimes	Always
Forgetting	Manual	Manual	Automatic (7d)
Associative Recall	No	No	Yes
Cost at Scale	Linear	Linear	Sublinear

Market position: Not an upgrade to existing memory systems. A new category.

Traction & MVP

Launch Status

- ✓ **Core CA engine:** 19 modules, production-ready
- ✓ **Automated testing:** 167+ tests across datasets (MBPP, SWE-bench, BABILong)
- ✓ **Web UI:** Wheeler dashboard with live recall, temperature display
- ✓ **Darman chatbot:** Live demo, integrated with Ollama
- ✓ **GPU support:** HIP/ROCm + CUDA with CPU fallback
- ✓ **Open source:** GitHub live, CC BY-NC 4.0, community-ready

No VC funding yet. Built by founding team. Ready to accelerate.

Named After Wheeler's "It from Bit"

"Every 'it' — every physical thing — derives its existence as an information system from answered physical questions."

— John Archibald Wheeler

Insight: In our system, meaning emerges from dynamics. Information isn't stored; it's *computed*. Memories exist because the CA *interacts* with them through repeated recall.

Philosophy: Information emerges from physical process, not from static representation.

Open Source, Commercial Ready

GitHub: `fantomx42/wheeler-memory`

- **License:** CC BY-NC 4.0 (open source, non-commercial)
- **Quick start:** 3 commands to running instance
- **Documentation:** Architecture, CLI, API, design principles
- **Community:** Issues, discussions, contribution guidelines
- **Integration ecosystem:** Ollama, Open WebUI, Hugging Face

Path to commercialization: Dual licensing, managed service, enterprise integrations.

Closing: "The Formula Is the Foundation"

Call to Action

We've built the engine. The market needs it. Let's scale it.

- **For investors:** Memory is the missing layer in the \$100B LLM market
- **For enterprises:** Privacy-first, on-device, auditable confidence signals
- **For the market:** A new category, not an incremental improvement

Next steps:

1. Seed round: 18-month runway to commercial product
2. Enterprise partnerships: First 3 pilot customers
3. Open source acceleration: Grow contributor base
4. Managed service: Wheeler-as-a-Service for API consumers

Demo + Questions

Try it now:

```
git clone https://github.com/fantomx42/wheeler-memory.git
pip install -e .
wheeler-ui
# Open http://localhost:7437
```

Contact: [Your email / website]

